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PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS HACKS STANDINGS CUSTOM INVOCATION

E. Idiot First Search

time limit per test: 2 seconds
memory limit per test: 512 megabytes

There is a binary tree of $n+1$ vertices (n **is odd**), with vertices labeled $0, 1, \ldots, n$. **At most one letter** can be written on each vertex of the tree, and all vertices initially have nothing written on them. The root of the tree is vertex 0.

In the tree, vertex 0 is the parent of vertex 1, while all other vertices have either 2 children or 0 children.

Bob is lost in one vertex of the tree and wishes to escape the tree by reaching vertex 0. This is very easy for most people with common sense. However, since Bob is an idiot, he created a new way of traversing the tree; introducing the "Idiot First Search".

When Bob is on vertex v ($1 \le v \le n$), Bob's movement is determined as follows:

- If vertex v is a leaf, Bob **always** moves to the parent of v; otherwise, check the next few conditions.
- If nothing is written on vertex v, Bob writes 'L' on vertex v and moves to the left child of v;
- If 'L' is written on vertex v, Bob **overwrites it** to 'R' and moves to the right child of v;
- If 'R' is written on vertex v, Bob **erases it** and moves to the parent of v.

It takes exactly 1 second for Bob to move to an adjacent vertex, so Bob will take exactly x seconds to perform x moves.

It has been shown that regardless of which vertex Bob starts on, Bob can reach vertex 0 in a finite (though possibly inexplicably large) amount of time. We don't know who proved it; surely it can't be Bob, but it is definitely proven.

For each vertex $k = 1, 2, \ldots, n$, please determine the total time it takes to reach vertex 0 if Bob started on vertex k, in seconds. As the values may be huge, you are only asked to compute them modulo $10^9 + 7$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 300\,001$, n **is odd**).

Each of the next n lines contains two integers l_i and r_i denoting the children of vertex i ($0 \le l_i, r_i \le n$).

For each vertex, $l_i = r_i = 0$ is given if the vertex has no children. Otherwise, l_i and r_i are the left and right children of vertex i.

It is guaranteed that the input defines a valid binary tree satisfying the conditions given in the statement.

It is guaranteed that the sum of n over all test cases does not exceed $300\,001$.

Output

For each test case, output n integers $\tau_1, \tau_2, \ldots, \tau_n$ separated by spaces.

Here, τ_k denotes the total time it takes to reach vertex 0 if Bob started on vertex k, modulo $10^9 + 7$.

Example

Codeforces Round 1080 (Div. 3)
Finished
Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msy:
Choose file: Choose File no file selected
Submit

→ Problem tags

dfs and similar dp trees
No tag edit access

→ Contest materials

- Announcement (en)
- Tutorial (en)

input Copy

```
3
1
0 0
5
2 3
0 0
4 5
0 0
0 0
7
2 3
4 5
0 0
6 7
0 0
0 0
0 0
```

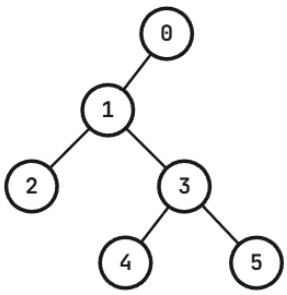
output Copy

```
1
9 10 14 15 15
13 22 14 27 23 28 28
```

Note

On the first test case, there are only two vertices, vertex 0 and vertex 1. It takes only 1 second for Bob to reach vertex 0 from vertex 1.

On the second test case, the tree is given as follows.



It takes 14 seconds for Bob to reach vertex 0 from vertex 3. The moves are as follows:

$$3 \xrightarrow{\texttt{L}} 4 \xrightarrow{\texttt{X}} 3 \xrightarrow{\texttt{R}} 5 \xrightarrow{\texttt{X}} 3 \xrightarrow{\texttt{X}} 1 \xrightarrow{\texttt{L}} 2 \xrightarrow{\texttt{X}} 1 \xrightarrow{\texttt{R}} 3 \xrightarrow{\texttt{L}} 4 \xrightarrow{\texttt{X}} 3 \xrightarrow{\texttt{R}} 5 \xrightarrow{\texttt{X}} 3 \xrightarrow{\texttt{X}} 1 \xrightarrow{\texttt{X}} 0$$

Here, the letters above the arrows denote the letter on the vertex before moving to the adjacent vertex, where $\texttt{X}$ denotes nothing written.

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